

SBI NEXUS



SBI NEXUS

AI-Powered Proactive Banking Relationship Platform

Banking that understands, predicts, and engages.



PRESENTED BY

Mohan Sai Pandeti

HACKATHON THEME

Agentic AI & Emerging Tech

Pillar 03: Digital Engagement

**AI
POWER**

SBI NEXUS



“Traditional banking models fail to proactively engage customers.”

- THE ENGAGEMENT GAP

REACTIVE NATURE

The current banking landscape relies heavily on reactive models where customers must initiate actions, explore services, and monitor their own spending.

MISSED OPPORTUNITIES

This creates significant missed opportunities, as customers frequently miss out on potential savings, timely investments, and critical bill payments due to a lack of real-time, context-aware guidance.

ENGAGEMENT GAP

Without personalized financial planning, the relationship between the bank and the customer remains strictly transactional rather than functioning as an advisory partnership.

01 / COMMUNICATION

INFORMATION OVERLOAD

Customers are overwhelmed by generic financial products that may not fit their current life stage or financial situation.

02 / SUPPORT SCALING

HIGH SUPPORT WORKLOADS

Banks struggle with high support workloads, facing high volumes of basic inquiries that could be easily resolved through intelligent, proactive communication.

03 / FRAUD PREVENTION

DELAYED FRAUD AWARENESS

Reactive alert systems often notify customers of unusual behavior after the fact, rather than proactively flagging risky patterns in real-time.



AI
POWER



SBI NEXUS • THE FUTURE OF DIGITAL ENGAGEMENT

SHIFTING BANKING FROM REACTIVE TO PROACTIVE

01 / INTELLIGENT COMPANION

AI-Powered Platform

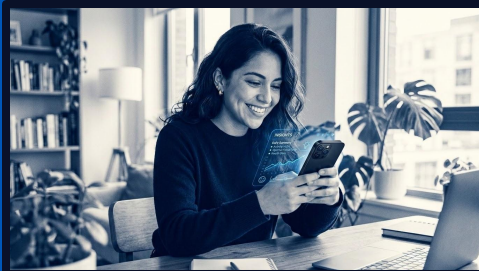
Introducing **SBI NEXUS**: an AI-powered, proactive digital engagement platform acting as an intelligent financial companion.

- **Always-On:** Operates seamlessly in the background to monitor spending patterns and goals.
- **Personalized:** Delivers smart recommendations right when they are needed most.

02 / PROACTIVE ENGAGEMENT

Continuous Analysis

NEXUS leverages continuous behavioral analysis to monitor customer spending patterns, goals, and banking activities in the background. This enables proactive engagement, shifting the paradigm from waiting for user prompts to delivering personalized, real-time recommendations directly within the existing SBI digital ecosystem.



AI
POWER





01 / SMART FORECASTS

Spending & Bill Prediction

SBI NEXUS forecasts recurring expenses and alerts users before upcoming due dates with highly accurate bill prediction algorithms.

02 / TARGETED SIP & LOANS

Product Recommendations

Suggests targeted banking products (SIPs, FDs, and Loans) exactly when customer financial behaviors signal an active need.

03 / MULTILINGUAL AI

Conversational AI

Breaks down language barriers by communicating financial insights dynamically in multiple Indian languages.



04 / FRAUD DETECTION

Proactive Security

Instantly flags anomalous financial behavior to increase customer security awareness and proactively stop unauthorized activity.

System Architecture

A robust architecture integrating Spring Boot, PostgreSQL, and Gemini AI to power the proactive customer dashboard.



01 / UI Layer

React UI

React Web and Mobile interface providing the primary frontend environment for the user.



02 / Backend

Spring Boot

Core backend utilizing Spring Boot with JWT Authentication for secure and reliable access.



03 / Data Layer

PostgreSQL

Robust data layer managing transaction histories and core customer profile records.



04 / AI & Gen AI Layer

AI & Gemini API

Recommendation Engine

Calculates financial health scoring, analyzes spending, and dictates product recommendations.

Generative AI

Gemini API integration generates deep, highly personalized, context-aware banking insights.



05 / Output

Dashboard

Customer Dashboard rendering real-time, proactive, and actionable financial alerts.

SBI NEXUS

SBI NEXUS TECH STACK

Built on a modern, scalable stack utilizing React, Spring Boot, and advanced AI APIs



FRONTEND

REACT.JS

TAILWIND CSS

BACKEND & APIs

SPRING BOOT

REST APIS

DATABASE

POSTGRESQL

AI & MACHINE LEARNING

GEMINI API

SPRING AI

RULES ENGINE

CLOUD & DEPLOYMENT

DOCKER

RENDER

RAILWAY

DEVELOPMENT TOOLS

GIT/GITHUB

POSTMAN

IDE / VS CODE



THE VISION NEXUS transforms the user experience from a contextual login to the seamless execution of proactive financial advice.

01**CONTEXTUAL LOGIN**

When a customer opens the SBI app, NEXUS immediately presents a customized dashboard based on recent activity.

02**PROACTIVE ALERT**

Notifies the user: "**You have ₹15,000 idle in your account after upcoming bills. Would you like to start a short-term FD?**"

03**CONVERSATIONAL ENGAGEMENT**

The user can then engage in Conversational Engagement, asking follow-up questions in their native language.

04**SEAMLESS EXECUTION**

Allows instant transaction approval, converting idle balances into active investments.

SBI NEXUS

ENTERPRISE AI LAYER

DIRECT INTEGRATION

NO NEW APP REQUIRED

SBI NEXUS operates as an enterprise AI engagement layer with direct integration across SBI's existing digital channels, driving immediate customer value without downloading friction.

REVENUE GENERATION



Drives higher adoption and cross-selling of loans, insurance, and investment products via timely, AI-driven nudges.

COST REDUCTION



Lowers operational overhead and substantially reduces customer support load through efficient automation.

CUSTOMER LOYALTY



Builds high customer loyalty, significantly improving customer satisfaction and driving record Net Promoter Scores.



01

Enterprise Cloud Deployment

Scaling the architecture using AWS or Google Cloud to support millions of concurrent users.

02

Advanced Wealth Management

Utilizing deeper predictive modeling for automated portfolio rebalancing and long-term wealth generation.

03

Agentic Autonomous Actions

Allowing users to grant the AI permission to automatically move small amounts to savings based on their daily micro-spending habits.

**VISION & EXECUTION**

**NEXUS IS BUILT TO SCALE SECURELY &
GENERATE WEALTH AUTONOMOUSLY**

SBI NEXUS



MEET THE DEVELOPER

FULL STACK

AI / ML

DATA SCIENCE



DEVELOPER

Mohan Sai Pandeti

Full Stack & AI Developer

Individual Participant

UNIVERSITY

Mohan Babu University

EDUCATION

CSE Data Science

Class of 2022 - 2026

ROLL NO

22102A030085